

Incident Response Analysis & Post-Incident Review: ICMP Flood Denial of Service (DoS)

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Governance Framework: NIST Cybersecurity Framework (NIST CSF 2.0)

Target Environment: Corporate Enterprise Network & Internal Services Infrastructure

Incident Classification: Availability Incident / Distributed/Volumetric DoS (MITRE ATT&CK T1498.001)

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1.0 Executive Summary & Event Timeline

1.1 Executive Summary

During normal operational hours, the organization's enterprise network experienced an unmitigated Denial of Service (DoS) attack, causing a complete operational outage of internal network services lasting approximately two hours. A threat actor exploited an unhardened, misconfigured edge firewall to flood network interfaces with excessive Internet Control Message Protocol (ICMP) echo-request packets.

This high-volume packet flood saturated physical bandwidth and interface buffers, severing access to shared file systems, local development nodes, and internal messaging services. Incident response personnel successfully contained the attack by dynamically dropping incoming ICMP traffic, shedding non-critical service loads, and restoring core infrastructure. This report provides the technical post-mortem and structures future resilience against the core functions of the **NIST Cybersecurity Framework (CSF 2.0)**: *Identify, Protect, Detect, Respond, and Recover*.

```
14:18:32.192571 IP your.machine.52444 > dns.google.domain: 35084+ A?
yummyrecipesforme.com. (24)
14:18:32.204388 IP dns.google.domain > your.machine.52444: 35084 1/0/0 A
203.0.113.22 (40)

14:18:36.786501 IP your.machine.36086 > yummyrecipesforme.com.http: Flags
[S.], seq 2873951608, win 65495, options [mss 65495,sackOK,TS val 3302576859
ecr 0,nop,wscale 7], length 0
14:18:36.786517 IP yummyrecipesforme.com.http > your.machine.36086: Flags
[S.], seq 3984334959, ack 2873951609, win 65483, options [mss 65495,sackOK,TS
val 3302576859 ecr 3302576859,nop,wscale 7], length 0
14:18:36.786529 IP your.machine.36086 > yummyrecipesforme.com.http: Flags
[.], ack 1, win 512, options [nop,nop,TS val 3302576859 ecr 3302576859],
length 0
14:18:36.786589 IP your.machine.36086 > yummyrecipesforme.com.http: Flags
[P.], seq 1:74, ack 1, win 512, options [nop,nop,TS val 3302576859 ecr
3302576859], length 73: HTTP: GET / HTTP/1.1
14:18:36.786595 IP yummyrecipesforme.com.http > your.machine.36086: Flags
[.], ack 74, win 512, options [nop,nop,TS val 3302576859 ecr 3302576859],
length 0
```

Figure 1.0(tcpdump traffic log for the security event)

1.2 Event Timeline

- **14:18:32:** Volumetric ICMP flood breaches edge perimeter via unconfigured firewall.
- **14:18:36:** Complete network saturation; routers drop packets; internal workflows halt.
- **14:29:40:** Incident Response (IR) team activated; root cause identified as inbound ICMP flood.
- **14:45:00:** Containment executed: emergency border firewall rules drop inbound ICMP; non-essential services stopped.
- **15:00:33:** Core routing, DNS, and file systems restored; baseline traffic stability validated.

- **16:05:32:** All services operational; post-incident monitoring and hardening protocols engaged.

2.0 Threat Characterization & Incident Impact

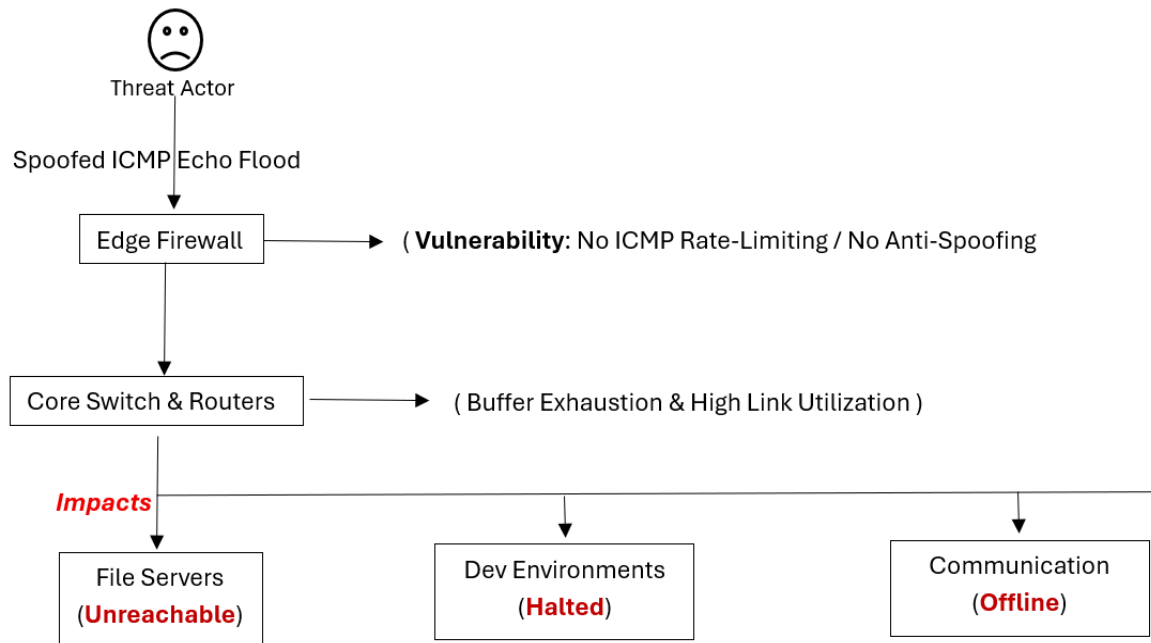


Figure 1.2(Attack Vector Schematic)

2.1 Attack Vector Mechanics

- **Vector Type:** ICMP Flood / Protocol Exhaustion (Network Layer).
- **Vulnerability Exploited:** Absence of perimeter rate-limiting thresholds and failure to enforce ingress packet validation rules on boundary firewalls.
- **Operational Impact:** Complete saturation of internal routing backbones; exhaustion of router state tables and interface queues; operational downtime totaling 120 minutes.
- **Impacted Assets:** Core routing equipment, internal file storage, local development nodes, and internal employee collaboration tools.

3.0 Alignment with NIST Cybersecurity Framework (CSF 2.0)

3.1 Identify (ID)

- **Root-Cause Analysis:** Telemetry analysis confirmed threat actor directed an ICMP Echo Request flood toward the enterprise edge gateway. The perimeter firewall operated with permissive default-allow policies for ICMP and lacked ingress inspection for spoofed RFC 1918 and external source addresses.
- **Asset Exposure Audit:** Audited all internal network infrastructure including edge routers, DNS endpoints, and production servers to confirm that no data breach, unauthorized privilege escalation, or secondary payload installation accompanied the flood event.

- **Vulnerability Mapping:** Classified the root weakness under CWE-400 (Uncontrolled Resource Consumption) and mapped the adversary behavior to MITRE ATT&CK Technique T1498.001 (Network Denial of Service: Direct Network Flood).

3.2 Protect (PR)

To shrink the external attack surface and ensure defensive depth, the following perimeter hardening controls were applied:

- **ICMP Rate-Limiting:** Implemented strict threshold controls at the edge firewall to drop non-standard ICMP traffic volumes before transit queues can saturate
- **Unicast Reverse Path Forwarding (uRPF) & Anti-Spoofing:** Enabled strict source IP address verification at the perimeter interface to evaluate and immediately drop spoofed source IP addresses (RFC 2827/BCP 38 compliance).
- **Attack Surface Reduction (Port Pruning):** Audited and closed all unneeded external entry ports, enforcing an explicit default-deny stance across all inbound boundaries.

3.3 Detect (DE)

To ensure immediate visibility and rapid identification of future anomalies, the detection stack was overhauled:

- **Network Behavioral Monitoring:** Deployed network monitoring agents using tools like tcpdump alongside continuous interface analytics to establish operational traffic baselines and generate automated threshold alerts for anomalous throughput spikes.
- **Intrusion Detection/Prevention Deployment (IDS/IPS):** Integrated Snort/Suricata sensor pipelines configured to identify and automatically drop volumetric ICMP attack patterns:
- **Centralized Firewall Log Ingestion:** Streamed border firewall connection state logs into a centralized log store to continuously monitor inbound traffic from untrusted external CIDR ranges.

3.4 Respond (RS)

In the event of active packet flooding or anomalous network behavior, standard operating procedures dictate:

- **Network Segmentation & Containment:** Dynamically isolate heavily affected network subnets to preserve the availability of unaffected core operational services and prevent lateral disruption.
- **Dynamic Access Control Updates:** Blackhole identified source addresses at the edge routing layer while maintaining valid, authenticated administrative connections.
- **SIEM Correlation & Forensic Ingestion:** Consolidate tcpdump packet captures, firewall access records, and host logs within Splunk to reconstruct attack timelines, establish source attribution, and refine firewall rules.
- **Out-of-Band (OOB) Incident Escalation:** Notify incident response leaders and technical system owners through out-of-band communication paths to maintain coordination during an enterprise network outage.

3.5 Recover (RC)

- **Tiered Production Restoration:** Restored business operations following a phased approach: core gateway routing and internal DNS first, operational data repositories and file servers second, and non-essential build/collaboration environments last.
- **Post-Restoration Packet Verification:** Maintained real-time capture filters via tcpdump to confirm traffic signatures had returned to baseline thresholds and ensure no secondary, low-and-slow exfiltration channels were active.
- **Configuration State Verification:** Verified firewall, routing table, and access-control list (ACL) persistence across all active gateways to confirm newly deployed rules would survive system reboots or cluster failovers.

4.0 Lessons Learned & Strategic Recommendations

Gap Identified	Immediate Remediation	Long-Term Strategic Control
Unrestricted edge ICMP ingress	Applied border ICMP rate-limiting and burst controls.	Formalize annual edge-firewall policy audits and continuous posture assessments.
Inability to detect spoofed packets	Configured strict source IP address verification at perimeter.	Implement BGP-level anti-spoofing and upstream DDoS scrubbing services.
Reactive alerting delays	Built real-time baseline threshold alerts via network monitors and IDS.	Integrate automated SOAR playbooks to drop offending CIDRs without manual analyst intervention.